

Supplementary material

Proteomics-based identification of biomarkers reflecting internal and external exposure to the advanced glycation end product precursor methylglyoxal in SH-SY5Y human neuroblastoma cells

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Table S1. Statistical results from the Jonckheere-Terpstra Test for trend analysis of total intensity of detected MGO-modified peptide(s) within each protein with increasing MGO concentrations (Protein IDs and Gene names were retrieved from the UniProt).

No.	Protein ID	Protein name	Gene name	J statistic ^a	p-value
1	P08670	Vimentin	VIM	4.40	0.00001
2	P68371	Tubulin beta-4B chain	TUBB4B	4.37	0.00001
3	Q5QNW6	Histone H2B type 2-F	H2BC18	4.08	0.00004
4	P26641	Elongation factor 1-gamma	EEF1G	4.05	0.00005
5	P60660	Myosin light polypeptide 6	MYL6	2.96	0.00310
6	P63261	Actin, cytoplasmic 2, N-terminally processed	ACTG1	2.29	0.02
7	P22732	Solute carrier family 2, facilitated glucose transporter member 5	SLC2A5	2.25	0.02
8	Q9Y2V2	Calcium-regulated heat-stable protein 1	CARHSP1	2.16	0.03
9	Q08378	Golgin subfamily A member 3	GOLGA3	1.70	0.09
10	Q6NSI1	Putative ankyrin repeat domain-containing protein 26-like protein	ANKRD26P1	1.56	0.12
11	Q9Y2F9	BTB/POZ domain-containing protein 3	BTBD3	1.51	0.13
12	Q9NXL9	DNA helicase MCM9	MCM9	1.25	0.21
13	Q8IVE3	Pleckstrin homology domain-containing family H member 2	PLEKHH2	1.05	0.29
14	Q8NCL9	Protein APCDD1-like	APCDD1L	0.98	0.33
15	Q86Y07	Serine/threonine-protein kinase VRK2	VRK2	0.89	0.37
16	P52298	Nuclear cap-binding protein subunit 2	NCBP2	0.88	0.38
17	Q9BXF9	Tektin-3	TEKT3	0.63	0.53
18	P60709	Actin, cytoplasmic 1, N-terminally processed	ACTB	0.63	0.53
19	A5A3E0	POTE ankyrin domain family member F	POTEF	0.63	0.53
20	P48643	T-complex protein 1 subunit epsilon	CCT5	0.57	0.57
21	P61550	Endogenous retrovirus group S71 member 1 Env polyprotein	ERVS71-1	0.37	0.71
22	P52790	Hexokinase-3	HK3	0.33	0.74
23	P61160	Actin-related protein 2	ACTR2	0.24	0.81
24	P52179	Myomesin-1	MYOM1	0.14	0.89
25	P09874	Poly [ADP-ribose] polymerase 1	PARP1	0.05	0.96
26	O75182	Paired amphipathic helix protein Sin3b	SIN3B	0.05	0.96
27	P43354	Nuclear receptor subfamily 4 group A member 2	NR4A2	0.05	0.96
28	Q9UJW0	Dynactin subunit 4	DCTN4	0.05	0.96
29	Q9H4B7	Tubulin beta-1 chain	TUBB1	0.00	1.00
30	Q9H3R1	Bifunctional heparan sulfate N-deacetylase/N-sulfotransferase 4	NDST4	-0.05	0.96
31	Q2VY69	Zinc finger protein 284	ZNF284	-0.05	0.96
32	O75541	Zinc finger protein 821	ZNF821	-0.10	0.92
33	Q96QB1	Rho GTPase-activating protein 7	DLC1	-0.17	0.86
34	Q9NWS8	Required for meiotic nuclear division protein 1 homolog	RMND1	-0.19	0.85
35	Q8N8I0	Sterile alpha motif domain containing 12.	SAMD12	-0.19	0.85
36	P51553	Isocitrate dehydrogenase [NAD] subunit gamma, mitochondrial	IDH3G	-0.20	0.85
37	P52789	Hexokinase-2	HK2	-0.23	0.81
38	Q9H069	Dynein regulatory complex subunit 3	DRC3	-0.24	0.81
39	P56524	Histone deacetylase 4	HDAC4	-0.28	0.78
40	Q96AA8	Janus kinase and microtubule interacting protein 2	JAKMIP2	-0.38	0.70

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Table S1 (continued)

No.	Protein ID	Protein name	Gene name	J statistic ^a	p-value
41	P13639	Elongation factor 2	EEF2	-0.47	0.64
42	P60510	Serine/threonine-protein phosphatase 4 catalytic subunit	PPP4C	-0.56	0.58
43	Q9NWB1	RNA binding protein fox-1 homolog 1	RBFOX1	-0.56	0.57
44	P17038	Zinc finger protein 43	ZNF43	-0.70	0.48
45	Q5T7B8	Kinesin-like protein KIF24	KIF24	-0.72	0.47
46	A5PLL1	Ankyrin repeat domain 34B	ANKRD34B	-0.75	0.46
47	Q9H0K1	Serine/threonine-protein kinase SIK2	SIK2	-0.75	0.46
48	Q8IZH2	5'-3' exoribonuclease 1	XRN1	-0.80	0.43
49	Q14683	Structural maintenance of chromosomes protein 1A	SMC1A	-0.89	0.37
50	Q92576	PHD finger protein 3.	PHF3	-0.89	0.37
51	Q4J6C6	Prolyl endopeptidase-like	PREPL	-0.93	0.35
52	P16152	Carbonyl reductase [NADPH] 1	CBR1	-0.99	0.32
53	Q00872	Myosin-binding protein C, slow-type	MYBPC1	-1.08	0.28
54	Q86VV8	Rotatin	RTTN	-1.08	0.28
55	P52566	Rho GDP-dissociation inhibitor 2	ARHGDI2	-1.13	0.26
56	Q9HCD5	Nuclear receptor coactivator 5	NCOA5	-1.17	0.24
57	Q13751	Laminin subunit beta-3	LAMB3	-1.27	0.20
58	O94906	Pre-mRNA-processing factor 6	PRPF6	-1.30	0.19
59	P10827	Thyroid hormone receptor alpha	THRA	-1.31	0.19
60	Q70UQ0	IKBKB interacting protein.	IKBIP	-1.50	0.13
61	Q8IXZ2	Zinc finger CCH domain-containing protein 3	ZC3H3	-1.54	0.12
62	Q9UGP8	Translocation protein SEC63 homolog	SEC63	-1.61	0.11
63	P35557	Hexokinase-4	GCK	-1.65	0.10
64	A8MPP1	Putative ATP-dependent RNA helicase DDX11-like protein 8	DDX11L8	-1.68	0.09
65	Q9Y4X0	AMMECR nuclear protein 1.	AMMECR1	-1.73	0.08
66	P25874	Mitochondrial brown fat uncoupling protein 1	UCP1	-1.78	0.08
67	P07355	AMMECR nuclear protein 1.	ANXA2	-1.83	0.07
68	A6NLP5	Tetratricopeptide repeat domain 36.	TTC36	-1.86	0.06
69	O14827	Ras-specific guanine nucleotide-releasing factor 2	RASGRF2	-1.88	0.06
70	Q8TED4	Glucose-6-phosphate exchanger SLC37A2	SLC37A2	-1.91	0.06
71	O15397	Importin-8	IPO8	-1.94	0.05
72	Q9H3R0	Lysine-specific demethylase 4C	KDM4C	-2.02	0.04
73	Q8TB72	Pumilio homolog 2	PUM2	-2.15	0.03
74	Q8N3X1	Formin binding protein 4.	FNBP4	-2.20	0.03
75	Q8N8K9	Uncharacterized protein KIAA1958	KIAA1958	-2.52	0.01
76	Q13938	Calcyphosin	CAPS	-2.80	0.01
77	O60304	Zinc finger protein 500	ZNF500	-2.87	0.004

^a The J statistic values are ranked in descending order and reflect the direction and magnitude of the trend in intensity of the modified peptides with increasing MGO concentrations. A positive J statistic value indicates an upward trend, suggesting that the intensity of protein modification increases as MGO concentration rises. A negative J statistic value indicates a downward trend, suggesting a decrease in protein modification intensity with increasing MGO concentrations.

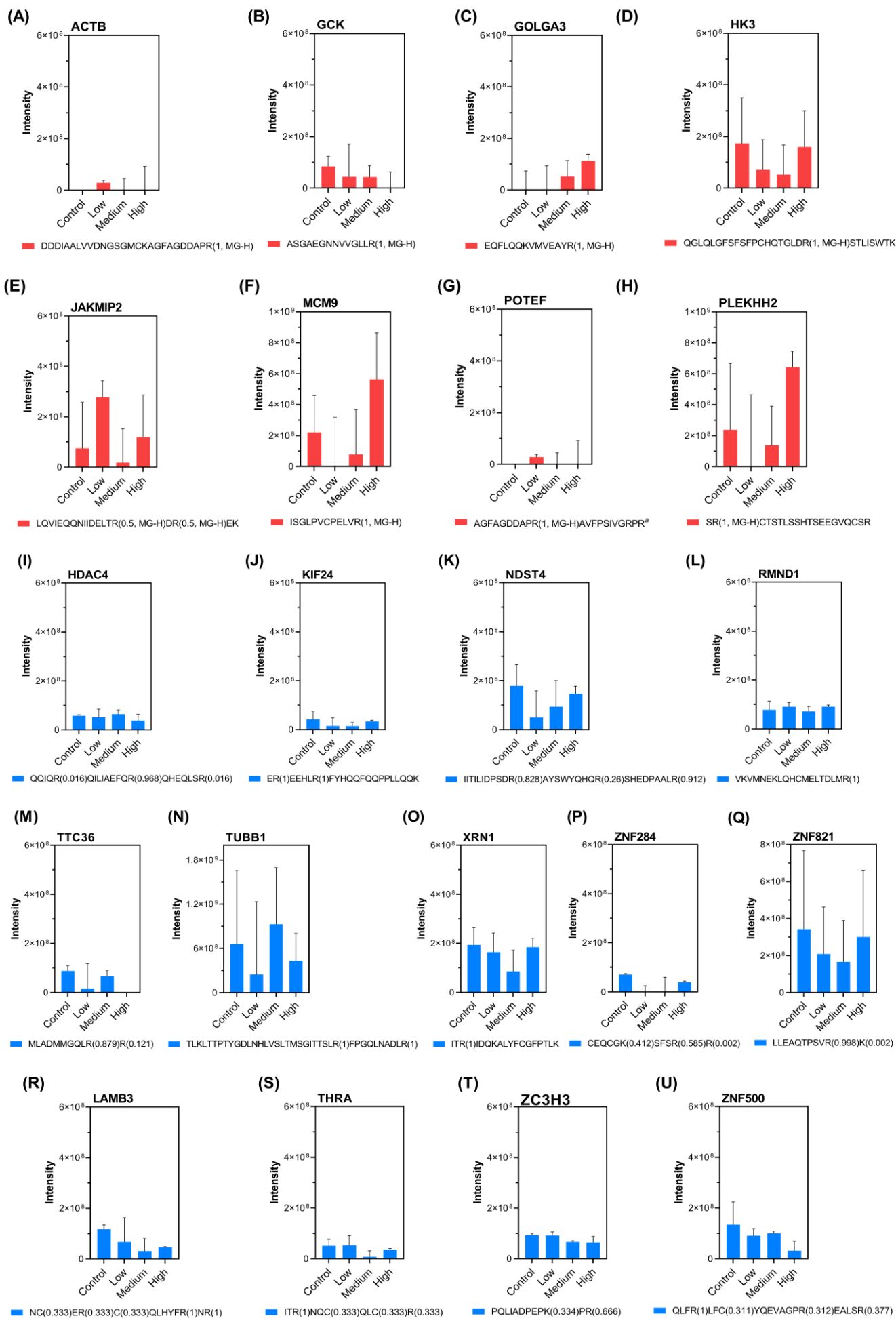


Fig. S1 (Continued)

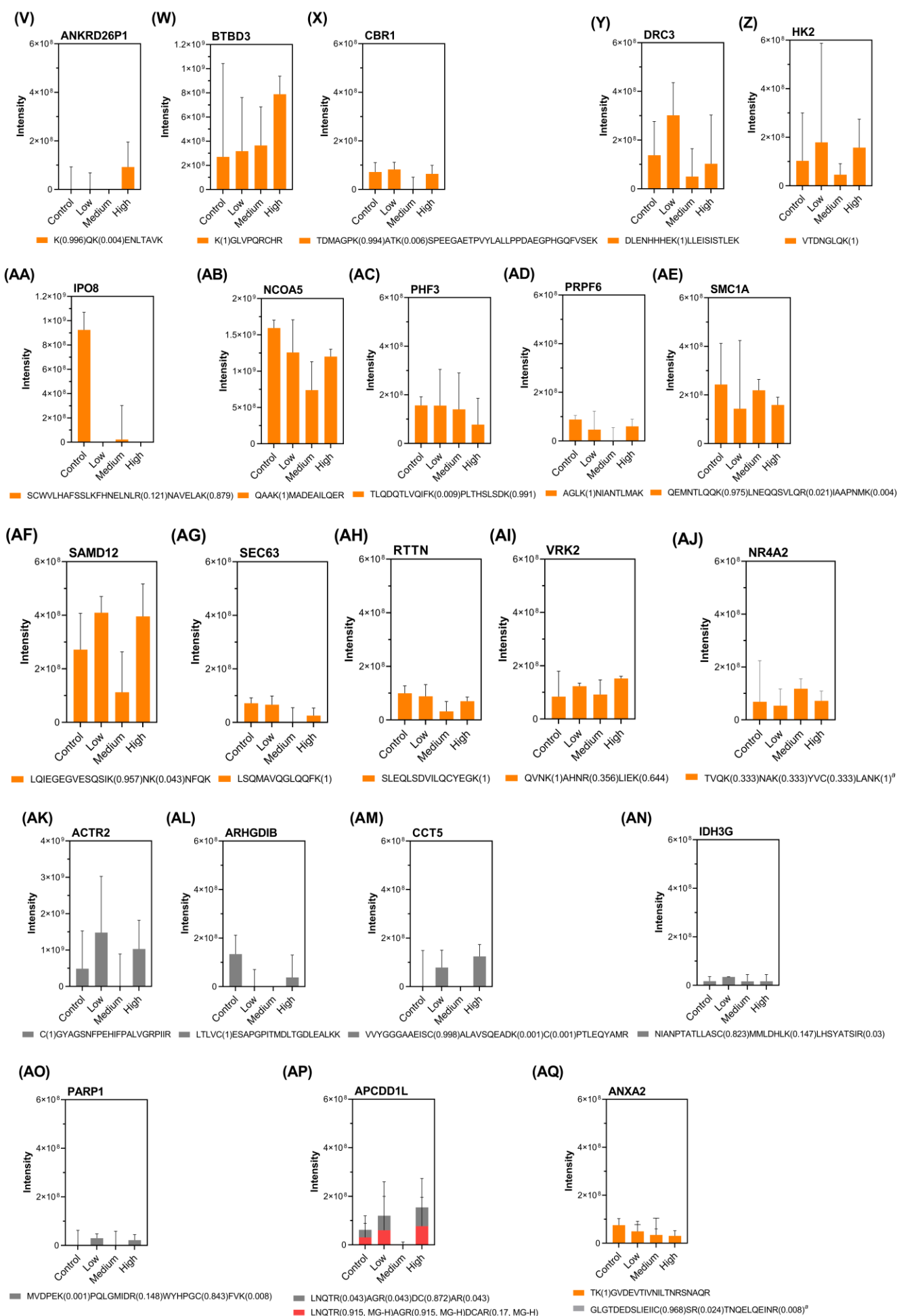


Fig. S1 (Continued)

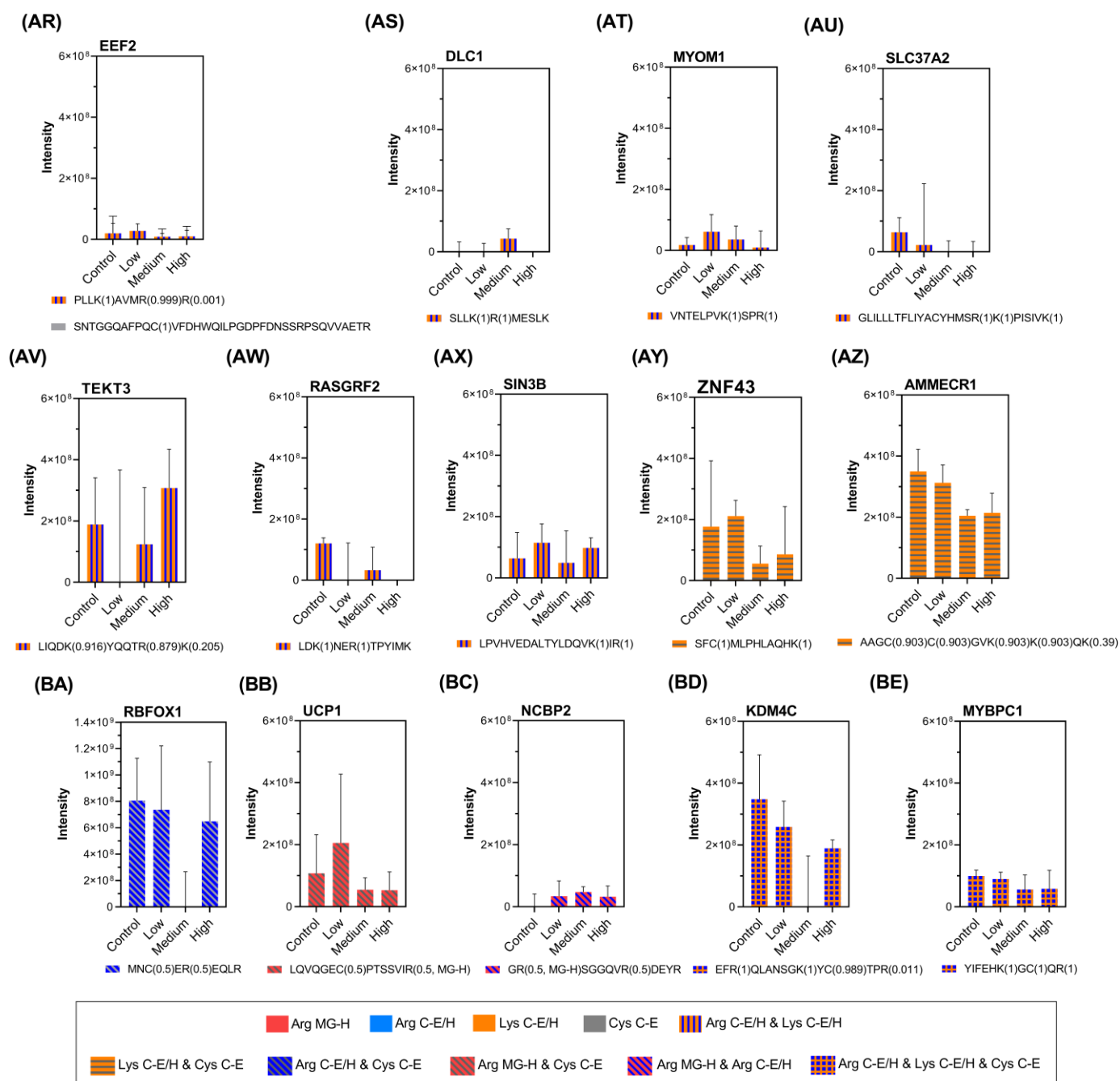


Fig. S1. Intensity profiles of detected MGO-modified peptides in proteins defined as highly susceptible MGO modification but not meeting biomarker selection criteria. “a” indicates peptide hits that also match additional homologous protein sequences. Numbers in parentheses next to amino acids represent the probability of modification at that site. “MG-H” is specifically annotated in parentheses to distinguish between the two types of arginine modifications being studied. Peptides with multiple potential modification types (each having a probability ≥ 0.5) are denoted by pattern fills as indicated in the figure legend. Data are presented as medians with interquartile ranges, derived from four independent replicates. Note the different Y axis in panel F, H, N, Q, W, AA, AB, AK, and BA as compared to other panels.